(2) a) $\lim\limits_{n\to\infty} \frac{n}{\sqrt{10+n}} = \lim\limits_{n\to\infty} \frac{n^2}{10+n} = \frac{n^2}{n\cdot\left(\frac{10}{n}+1\right)} = \frac{n}{\cancel{\frac{10}{n}}+1} = \infty$ ∴ diverge

b) $\lim\limits_{n\to\infty} \frac{5}{n+2} = 0$ ∴ converge

(3) a) * Teste da razão $\lim\limits_{n\to\infty} \sqrt{\frac{1}{n\cdot(\ln(n))^2}} = 0$ $0<1$ ∴ A série converge

c) * teste da raiz: $\lim\limits_{n\to\infty} \sqrt[n]{\frac{n^2}{e^n}} = \lim\limits_{n\to\infty} \frac{\sqrt[n]{n^2}}{e} = 0$ $0<1$ ∴ converge

b) * Comparação de limites:

$b_n = \frac{1}{n^3} \to$ diverge $L<0$ ∴ Inconclusivo

$\lim\limits_{n\to\infty} \frac{\frac{5-2\sqrt{n}}{n^3}}{\frac{1}{n^3}} = \lim\limits_{n\to\infty} 5-2\sqrt{n} = -\infty$

* Teste da comparação:

d) * teste da razão: $\lim\limits_{n\to\infty} \frac{\frac{\cancel{(n+1)}\cdot\cancel{7^n}\cdot 7}{\cancel{(n+1)}\cdot\cancel{n!}}}{\frac{n\cdot\cancel{7^n}}{\cancel{n!}}} = \lim\limits_{n\to\infty} \frac{7}{n} = 0$ $0<1$ ∴ converge

(4) Parte fixa: $\frac{123}{10^3}$

Parte periódica: $\frac{456}{10^6} + \frac{456}{10^9} + \frac{456}{10^{12}} \cdots$ PG $r = \frac{1}{10^3}$ $S = \frac{\frac{456}{10^6}}{1-\frac{1}{10^3}} = \frac{\frac{456}{10^6}}{\frac{1000-1}{10^3}} = \frac{456}{999000}$

Total: $\frac{123}{1000} + \frac{456}{999000} = \frac{123333}{999000}$ //

(5) a) * Teste da raiz $\to \lim\limits_{n\to\infty} \sqrt[n]{\frac{1}{2^n}} = \lim\limits_{n\to\infty} \frac{1}{2} = \frac{1}{2}$ $\frac{1}{2}<1$ ∴ converge

$\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16} \cdots\right)$ $S = \frac{a_1}{1-r} \to S = \frac{\frac{1}{2}}{1-\frac{1}{2}} = \frac{\frac{1}{2}}{\frac{1}{2}} = 1$

A soma vai convergir para 1.

b) * Telescópica: $\frac{1}{n(n+3)} = \frac{A}{n} + \frac{B}{n+3}$ $A = \frac{1}{3}$ $B = -\frac{1}{3}$

$\frac{2}{3}\cdot\left(\frac{1}{n} - \frac{1}{n+3}\right) \to \frac{2}{3}\left[\left(1-\cancel{\tfrac{1}{4}}\right)+\left(\frac{1}{2}-\cancel{\tfrac{1}{5}}\right)+\left(\frac{1}{3}-\cancel{\tfrac{1}{6}}\right)+\left(\cancel{\tfrac{1}{4}}-\frac{1}{7}\right)\cdots\right]$

Soma $= \frac{2}{3}\cdot\left(1+\frac{1}{2}+\frac{1}{3}\right) = \frac{11}{9}$

(6) Se $\Sigma|a_n|$ converge, $\Sigma\, a_n$ converge. ∴ converge absolutamente

$\sum\limits_{n=1}^{\infty} \frac{1}{(n+1)^2} \to p>1$ ∴ converge.